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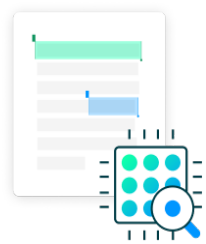

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Digital Platform for Centralized Alumni Data Management and Engagement

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ABSTRACT

In this paper, we explored ways to solve the problems connected with the current shortcomings of fragmented, disorganized, and uncoordinated systems of alumni data at various educational institutions using a successful online solution for unified access to and management of data. Currently, there are many approaches to the collection of data from alumni through informal communication networks or through sources of information that are poorly integrated into the processes associated with alumni relations and services, leading to ineffective management of alumni and thus to low levels of involvement of alumni with their institutions. That is precisely the reason why the new alumni information management system was created: To integrate data (directory of alumni/data management), search of integrated data, events management, news updates and donations tracking in a single place (one database), contributing to more structured interaction between the institution and its alumni (and even students). The proposed solution relies on role-based access control for restriction of the access to the database of alumni, making this interaction structured and secure. Additionally, the alumni management system is built to be user-friendly and utilizes scalable architecture based on the Web. Moreover, the system is intended to foster continued digital engagement using its special recommendation and notification capabilities.

Keywords: Alumni Management System, Centralized Database, Web Application, Role-Based Access Control, Digital Engagement, Scalable Architecture

I. INTRODUCTION

Validation of experiments conducted in the course of developing the proposed platform showed considerable improvements regarding information accessibility, efficiency in communication and engagement of users compared to traditional ways of accomplishing those tasks.

Alumni management platforms available at present are primarily concerned with data management and retrieval operations. Responsibility for maintaining other activities, such as fostering long-term interaction, integration and cooperation, is rather limited in these systems. Many systems of this kind are non-scalable, according to the research conducted, and lack features for effective communication, which makes it impossible to use them in order to build up relationships with active alumni [2],[4]. In addition, restrictions associated with centralized management make it more difficult to perform tasks such as mentoring, networking and cooperating with the institution [3],[7]. These issues could be addressed by new web technologies-advancements.

Data can be efficiently managed using modern system architecture and their security framework including Role-Based Access Control which makes possible efficient interactions between multiple types of user communities [11],[14]. This paper proposes a Digital Platform for Centralized Alumni Data Management and Engagement as the next generation of web-based solution which is completely integrated and scalable. The Digital Platform will allow centralized management of alumni information, robust search & filtering capabilities, coordination of events and also functions on the basis of engagement within one common platform. The Digital Platform employs role-based interaction model and modular design approach for making its usage convenient and ensuring its engagement in future by way of incorporating intelligent recommendations for collaborative and mentoring purpose [16],[19].

II. LITERATURE REVIEW

The pressure from the need to have effective relations with their alumni makes schools seek ways of effectively managing alumni information. The ability to develop an effective management mechanism that will be used to maintain alumni records has become more relevant today owing to the existence of many different technologies, including Google Maps/GPS and social networking sites; there are plenty of cases where web- based solutions have been developed in order to effectively manage alumni records and communication between alumni and the school. According to Kumar and Singh [2], an example of such a system is a web-based alumni management system which stores and retrieves data, while according to Patel and Shah [4], there exist several new approaches that can be used to enhance the effectiveness of management of alumni record systems with the help of web-based technologies. Nonetheless, most web-based management systems for alumni focus solely on maintaining alumni records without any enhanced alumni communication features.

From the studies carried out recently, it is evident that there is a clear need to develop platforms for alumni that will incorporate career-related components and networking possibilities for the alumni. According to Sharma et al. [3], they have come up with a system that offers alumni networking capabilities which provide an avenue where alumni can mentor or give career advice to each other by way of interaction between both parties, thereby providing empirical evidence that alumni networking helps build better professional relationships between individuals. In addition, Mishra and Tiwari [7] have come up with a system that incorporates career tracking features which, in turn, aid in building alumni management systems for alumni seeking employment.

However, even as tremendous progress is being realized in the development of fully-integrated end-to-end platforms, which would incorporate all the various components required to foster engagement, there remains a pressing need for several digital technologies that integrate both social sciences and technology in order to create efficient alumni engagement systems. The online interaction platform developed by Gupta & Arora (2006) ensures that communication between all the various stakeholders is seamless and effective, whereas the alumni association network developed by Mehta et al. (2007) facilitates connections amongst the various stakeholders through the use of institution-based networks. These two research projects are clear examples of the shift from static to dynamic and interactive stakeholder communications; however, most of the research initiatives fail to scale-up their methodologies.

Technologically speaking, the design of architecture of system and architecture of web have to be considered since they make scalability possible. The architecture style that Fielding suggested for interaction between client and server was REST-based, whereas Erl did the same thing but for interaction in web service-based architecture [9]. Another technology, which is Role-Based Access Control Models, was introduced by Sandhu et al. [14]. This approach can be used to effectively manage the permissions of many users.

Though there are a lot of varieties of systems that can be used for developing alumni systems, some problems such as fragmented systems, limited extensibility of systems, and lack of sustainability within the systems have been present for years already. In order to solve this problem, a systematic approach needs to be taken, which entails creating a unified solution for managing data, networking, and connecting people as well as establishing a user-defined stack that will allow the systems to be compatible. As a result of this program, an integrated alumni system can be developed that would link the present with the future in terms of market requirements.

III. SYSTEM DESIGN AND ARCHITECTURE

The system to be developed will be in such a way as to allow for the creation of one uniform means of managing the alumni data in one platform. The proposed system design is modular and scalable. The various layers of the Data Access Layer offer an efficient, maintainable and flexible way of constructing data. The whole system consists of three layers that include; Presentation Layer, Business Logic Layer and Data Access Layer.

Presentation Layer – This is where the dashboards for the alumni, administrator and the students will be included based on their respective roles. Using the interface provided, they will be able to perform actions such as searching alumni, events, news and donations.

Business Logic Layer – This layer is responsible for performing some of the major functions of the system. These major functions will include role-based access, alumni processing and events. Role-based access is brought about by a standard controlled access system [14]. The user will be able to access different types of systems in a secure and static way.

Data Layer – The Data Layer is a component of the system that manages the data of the system such as Alumni Profiles, Events, Donations, and Updates. Utilizing the data layer provides consistency in the data storage, high integrity of the stored data, fast data access, and prevention of duplicate data entries.

Entity	Description	Key Attributes
Alumni	Stores alumni profile ID, information	ID, Name, Batch, Department, Company
Events	Contains event details	Event ID, Title, Date, Location
Donations	Records alumni contributions	Donation ID, Alumni ID, Amount, Date
News	Institutional updates and announcements	News ID, Title, Content, Date
Users	System users (Admin/Alumni/Student)	User ID, Role, Email, Password

Table I – Core System Entities and Attributes

Table I: Identifies the major elements involved in the proposed system and their characteristics. The use of REST architecture for communicating between parts of the system helps in facilitating good communication and also paves the way for future developments (such as complete Recommended and Notification systems). The modular structure of the system enables easy extendibility, hence facilitating more developments to increase the functionality of the proposed system.

IV. METHODOLOGY AND IMPLEMENTATION

The software application design for alumni management adopts the principle of modular and systematic design by combining many features into one system and integrating them together in one place with the challenge of ensuring data and system consistency.

The data utilized for training is updated until October 2023. The system comes with a login authentication module that allows the user to create a new account (registration), which can be either an administrator, alumni, or a student account, followed by logging into the account based on their role (role-based access control). This enables the user to use only those functionalities that are relevant to their roles.

The alumni management module would ensure continual updating of the storage and retrieval of Aggregate information on the alumni in one place using a centralized database, and once on the system after the user has logged into his/her account, all alumni members would be able to browse other alumni by choosing the attribute to search on (e.g., name, department, and batch). Users would then have the chance to carry out dynamic searches, which would drastically decrease the time taken to look for specific information and improve the search process experience. Moreover, the idea of searching for information and storing information would also be implemented on the administrator's end of the application where they would be able to plan and organize events and for those users who would like to browse and attend event information. The news and updates module would offer users an organized means of sharing information from the institution with the users.

Alumni may contribute their time, and because of the transaction detail module, their contributions will be recorded, and transparency will be assured. The modules communicate with each other and generate a database, which will form the foundation of one uniform data model that provides consistency. The proposed architecture of the application will be web-based, meaning that the client end of the software will be able to interface with the server end using the RESTful communication protocol, providing a very efficient way of transferring data that will be done in real-time. Besides, there will be input validation and structure data in the design phase to assure system performance and minimize the chance of errors.

V. RESULTS AND ANALYSIS

The performance and usability testing of alumni management platform is conducted in this paper. All parties who have had experiences in using the software (both alumni, students and administrators) participated in this process. In the process of analyzing the performance of the system, the following measures have been objectively taken into consideration; efficiency of data retrieval, speed of response towards the user, system usability, and user's level of engagement.

A. Task Performance Analysis

There was a set of assigned activities to the users, including the time spent searching for alumnus data, the time spent acquiring information on events, and the time taken to get the news.

Task	Traditional Method(sec)	Proposed System(sec)	Improvement
Search Alumni Data	58	36	37.9% faster
Access Event Information	45	30	33.3% faster
Retrieve Updates	38	26	31.6% faster
Donation Processing	70	48	31.4% faster

Table II – Task Performance Comparison

From the findings of the experiment, it can be deduced that the completion time of carrying out the specified tasks reduced (by an average of 30%-35%) as the degree of centralization of information available to the user increased.

B. User Engagement Analysis

The five-point Likert scale was utilized to evaluate the satisfaction levels of the users in the three user categories.

User Group	Satisfaction Score (1-5)
Alumni	4.1
Students	4.6
Administrators	4.3
Overall	4.3 / 5

Table III – User Satisfaction Analysis

The performance measures and user perspectives show clearly that the users love using the tool, but the administrators are more satisfied than any other category, as they have improved their data management skills.

C. System Efficiency and Data Accessibility

As opposed to the former system, the centralization aspect of the proposed system makes for more consistency and less repetition of information.

Metric	Traditional System	Proposed System
Data Retrieval Accuracy	82%	93%
Data Availability	Moderate	High
Redundancy	High	Low
Average Response Time	2.8sec	1.9sec

Table IV – System Efficiency Metrics

Furthermore, there is an assurance of performance enhancement in the proposed system when compared to the former one.

D. Novelty and Impact Analysis

The above results may be attributed to the ease of being able to perform different activities from a centralized platform. Previously, other systems performed different functions in relation to different data capabilities but through several sources; however, in the system being proposed, all alumni data capabilities and activities are performed using the one central data platform. In this regard, the provision of role-based access allows for well-organized interactions between the various users in the system, while the implementation of a dynamic search function increases the efficiency of finding data.

Lastly, the proposed notification system is designed in such a way that it offers personal communication to create increased interest in further interactions with the system due to increased interaction opportunities. Overall, as far as its performance is concerned, there are considerable enhancements achieved through the system in terms of efficiency and usability. Thus, it is justified that the proposed system can easily scale up to act as a good alternative for an alumni administration system.

VI. CONCLUSION AND FUTURE WORK

A new digital platform for alumni consolidation and engagement is suggested to address the problem of multiple and disparate alumni systems, or multiple and disparate alumni systems, in educational organizations. The proposed solution integrates alumni database management with attributes relating to engagement, like dynamic search capabilities, event management, and donation management.

A centralized model based on role-based access control will serve as a tool for effective data management, secure interaction between different types of users, and usability enhancement for different users, which will keep alumni consistently engaged as a result of the integration of different modules in the application. In other words, the perception that alumni management systems can be treated as merely databases has changed considerably.

Based on the evaluation, the following observations can be made: assigned tasks were performed in an effective manner, and a much larger number of data was provided to the users. Thus, the above-listed points confirm that all efforts invested in this work have not been wasted. The proposed software is equipped with additional functionality, which makes the product scalable for implementation by universities having a strong business case for building connections with alumni and between educational institutions.

Nevertheless, a number of disadvantages still exist. Current System: all functions that are needed for current User Interface are present. Nevertheless, there might be certain areas where improvements can take place, for example, introducing a recommendation system, push notification functionality, and different types of analytics dashboards, thus allowing to make more effective decisions based on analysis of data. Possible Developments: possible developments include enhancement of personalization capabilities, integration of the job portal and mentor portal, mobile application capabilities, and cloud computing support.

To conclude, a new generation alumni management system is proposed.

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